Module 4 - Assignment 2 Power BI

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ALY6070: Communication & Visualization for Data Analytics

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Introduction

This report presents a data visualization analysis of the Telco Customer Churn dataset, originally published by IBM and sourced from Kaggle. The goal of this assignment is to design and build an interactive Power BI dashboard that effectively communicates insights about customer churn in a telecom company. The analysis begins with an exploration of the dataset structure, identifying key variables, patterns, and the intended audience. It then walks through each visualization, explaining why it was chosen and what insight it reveals. Finally, the report discusses how the dashboard brings all these elements together to tell a clear, data-driven story.

Analysis

Dataset Overview

The dataset contains 7,043 customer records across 21 variables. These variables can be categorised as follows.

Categorical variables include gender, SeniorCitizen, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies,Contract, PaperlessBilling, PaymentMethod , and Churn.

Discrete numerical variables include tenure, which represents the number of months a customer has been with the company, ranging from 1 to 72, and SeniorCitizen, which is coded as 0 or 1.

Continuous numerical variables include MonthlyCharges, which ranges from approximately $18 to $118, and TotalCharges, which captures the cumulative revenue from each customer over their tenure.

Calculated Fields:

1. **Churn Rate** - Calculated as the number of customers who churned (Churn = "Yes") divided by the total number of customers, multiplied by 100 to express it as a percentage.

2. **Avg Revenue Per Customer** - Calculated as the total charges across all customers divided by the total number of customers, giving the average revenue generated per customer.

Correlations, Patterns, and Trends

An initial exploration of the data reveals several notable patterns. Customers on month-to-month contracts show significantly higher churn compared to those on one-year or two-year contracts. Churn is heavily concentrated in the first few months of tenure, with a sharp decline as

customers stay longer. Fiber optic internet subscribers churn at a much higher rate than DSL or no-internet customers, despite paying higher monthly charges. Customers who pay via electronic check have a noticeably higher churn proportion than those using automatic payment methods. Gender, on the other hand, does not appear to be a meaningful predictor of churn, with both male and female customers showing nearly identical churn rates.

Audience and Key Questions

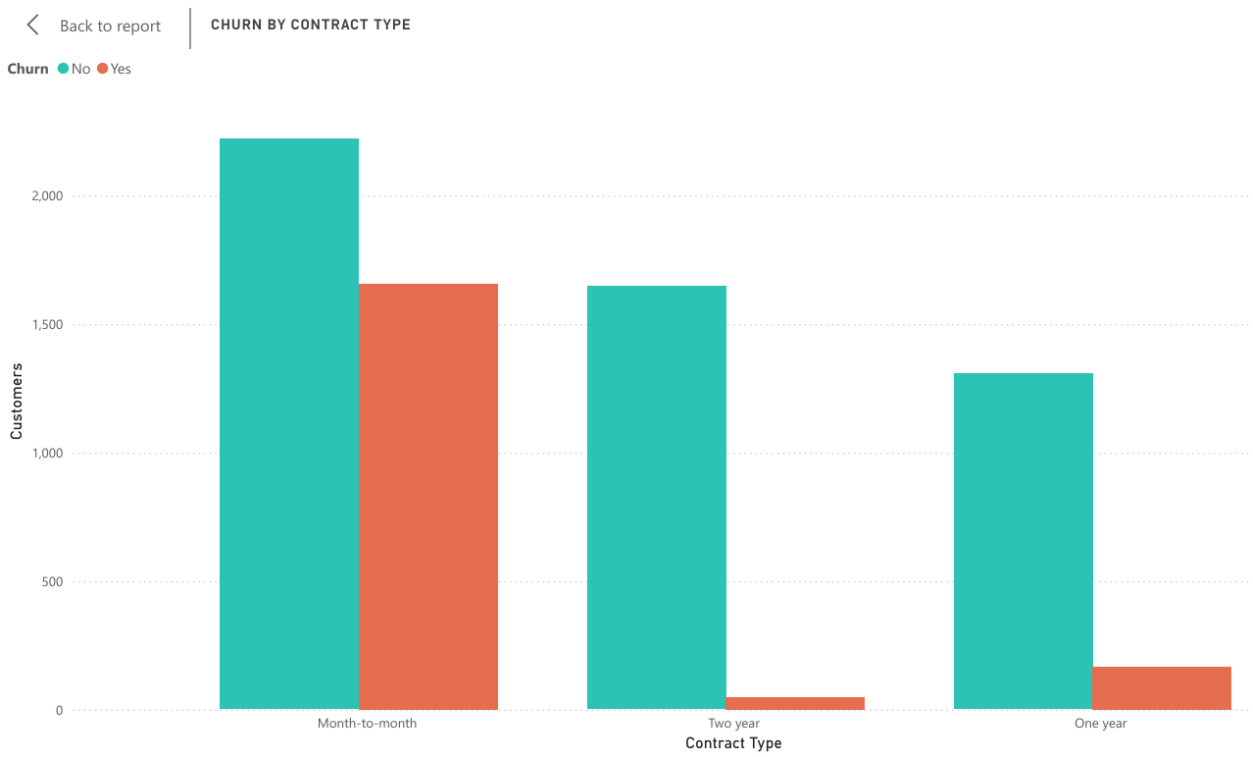
The intended audience for this dashboard includes telecom business managers, customer retention strategists, and CRM analysts. These stakeholders would typically ask questions such as: what is the overall churn rate and how many customers are we losing? Which contract types are most associated with churn? Does the type of internet service affect customer retention? Are certain payment methods linked to higher attrition? How quickly do customers tend to leave after joining? The dashboard is designed to answer all of these questions through a combination of KPI cards and interactive charts.

Visualizations

Each visualization in the dashboard was designed to answer a specific business question. The charts use a consistent teal and coral colour palette, where teal represents retained customers and coral represents churned customers, ensuring visual coherence across the dashboard.

Research Question 1: Which contract types are most associated with customer churn?

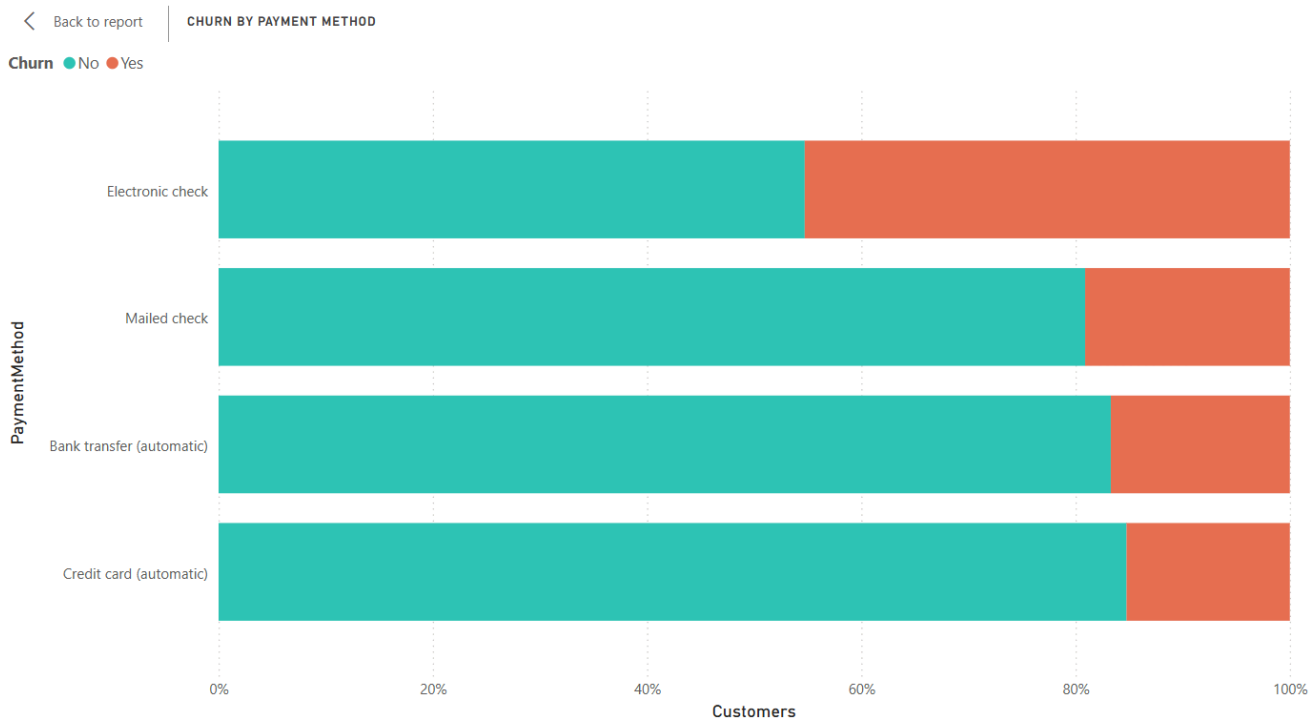
Figure 1: Churn by Contract Type



This clustered bar chart compares the count of retained and churned customers across three contract types: month-to-month, one year, and two year. A bar chart was selected because it is the most effective visual for comparing discrete categorical groups. The chart clearly shows that month-to-month contracts have the highest number of churned customers, while two-year and one-year contracts have significantly lower churn. This chart also serves as the main interactive filter for the dashboard - clicking on any contract type filters all other visuals, enabling the audience to explore churn patterns within a specific segment. This follows the gestalt principle of common fate, where linked elements respond together to user interaction.

Research Question 2: Does payment method influence churn behaviour?

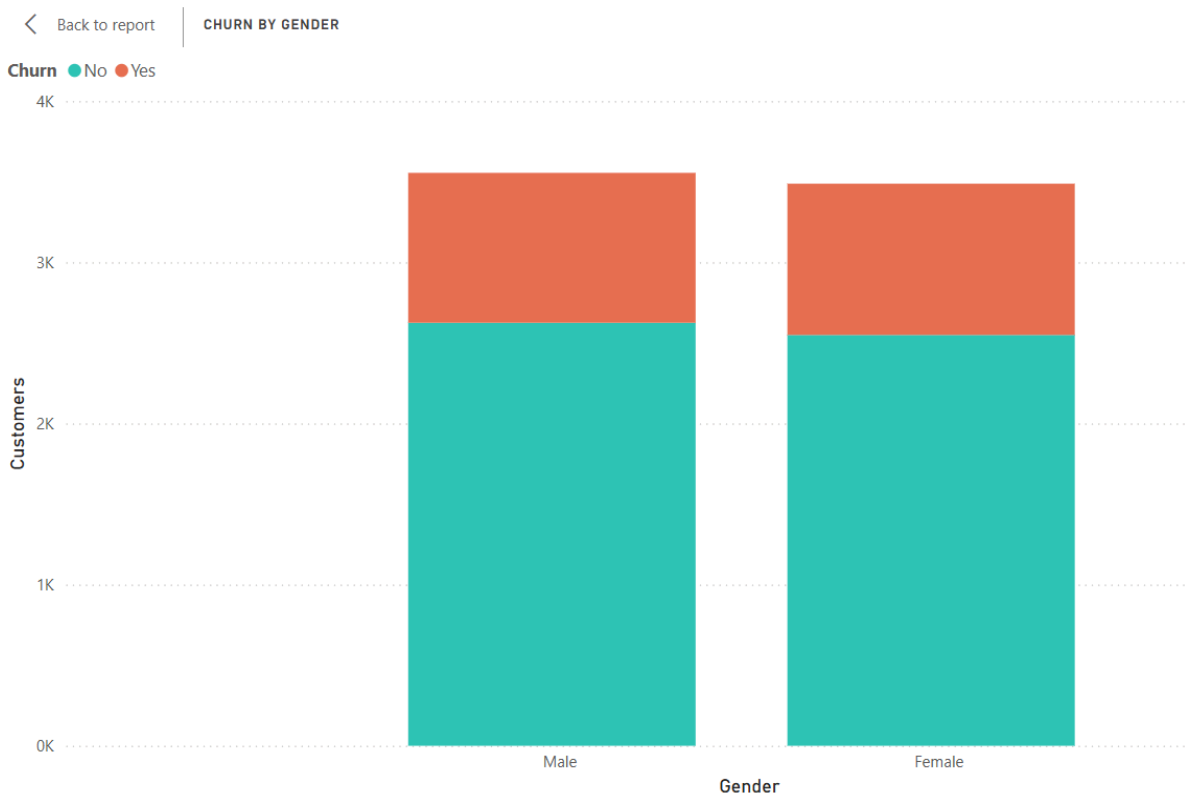
Figure 2: Churn by Payment Method



This 100% stacked horizontal bar chart displays the proportion of churned versus retained customers across four payment methods. A stacked bar was chosen to highlight proportional differences rather than absolute counts. The horizontal orientation ensures that the payment method labels remain fully readable without rotation, following the design principle of legibility. The chart reveals that electronic check users have a considerably higher churn proportion compared to customers using bank transfer, credit card, or mailed check. This suggests that customers on manual or less committed payment methods may be less engaged with the service.

Research Question 3: Is there a gender-based difference in churn?

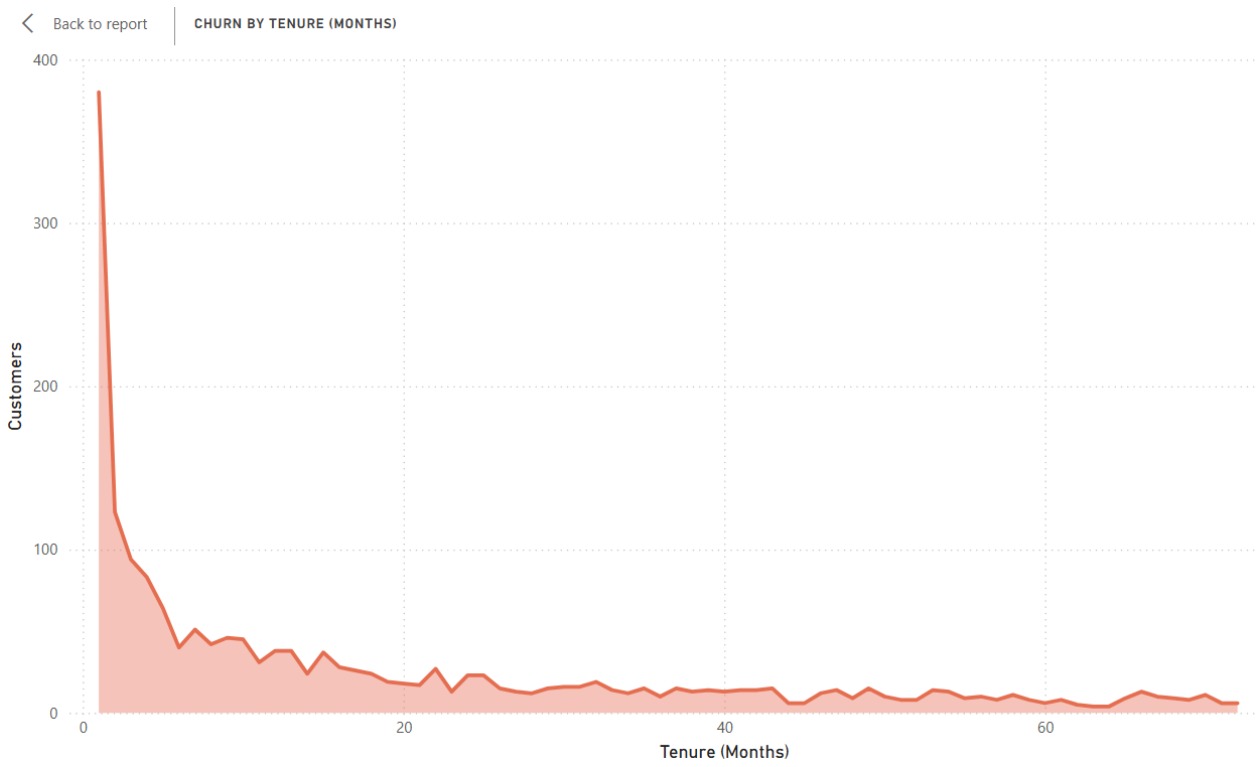
Figure 3: Churn by Gender



This clustered bar chart compares churn counts between male and female customers. A simple bar chart was appropriate here because there are only two categories to compare. The chart shows that churn is nearly identical across both genders, confirming that gender is not a significant driver of attrition. While this may seem like a non-finding, it is valuable for the audience because it helps retention teams avoid misdirecting resources toward gender-based segmentation strategies.

Research Question 4: When do customers tend to churn — early or late in their lifecycle?

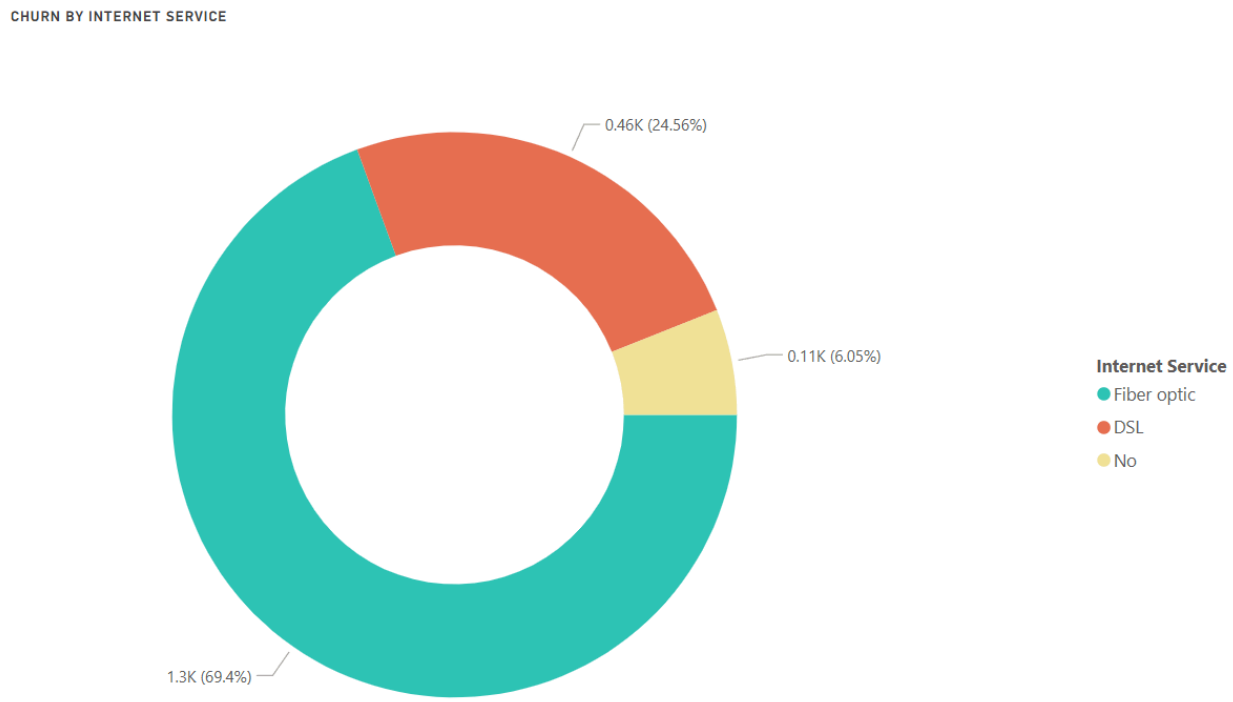
Figure 4: Churn by Tenure



This area chart plots the count of churned customers against their tenure in months. An area chart was chosen over a simple line chart because the filled area emphasises the volume of churn concentrated in the early months. The chart shows a dramatic spike in churn during the first one to five months, with a rapid decline as tenure increases. This pattern indicates that the onboarding phase is the most critical window for customer retention. Customers who survive the first few months are far more likely to stay long term.

Research Question 5: Which internet service type has the highest churn?

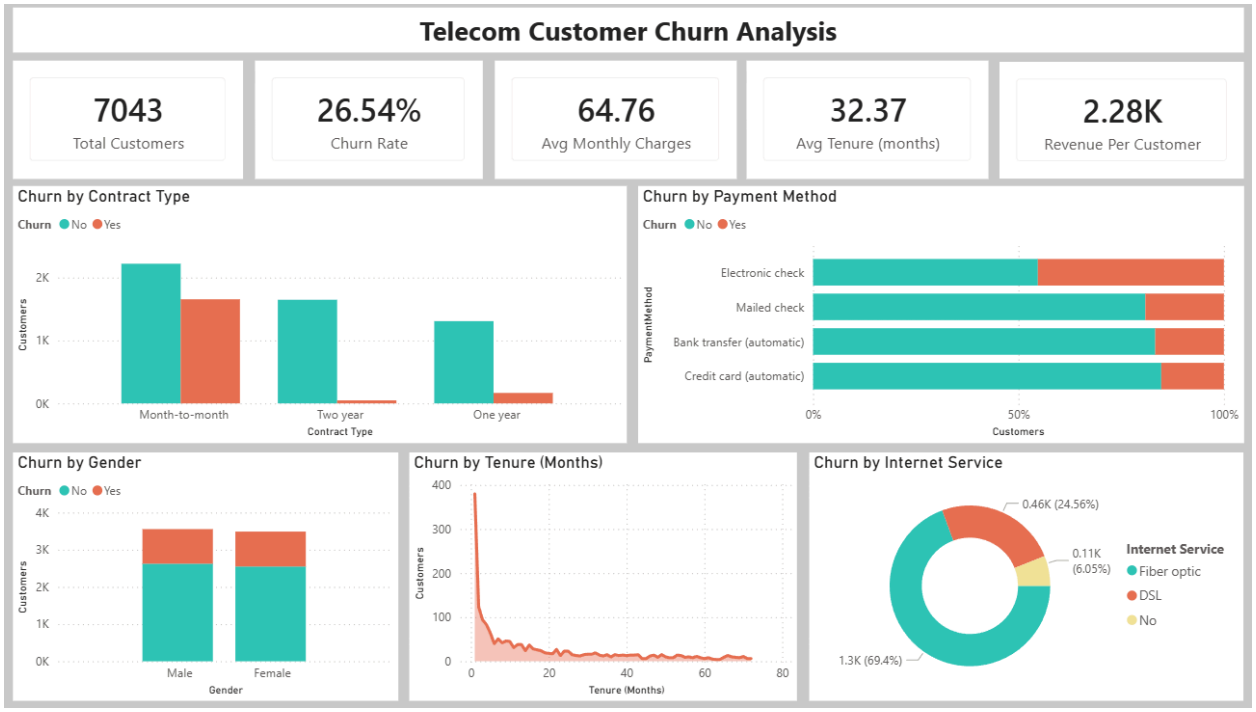
Figure 5: Churn by Internet Service



This donut chart displays the distribution of churned customers across three internet service types: Fiber optic, DSL, and No internet. A donut chart was selected because it effectively communicates part-to-whole relationships when the number of categories is small. The chart reveals that 69.4% of churned customers subscribed to fiber optic service, 24.56% to DSL, and only 6.05% had no internet service. This is a significant insight because fiber optic customers pay the highest monthly charges yet churn the most, suggesting potential service quality or expectation issues.

Dashboard Design

Figure 6: Telecom Customer Churn Analysis Dashboard



The dashboard brings all five visualizations together on a single page, along with five KPI cards at the top showing Total Customers (7,043), Churn Rate (26.54%), Average Monthly Charges ($64.76), Average Tenure (32.37 months), and Revenue Per Customer ($2.28K). Two of these KPI values, Churn Rate and Revenue Per Customer, are calculated fields created using DAX measures with DIVIDE and CALCULATE functions.

The layout follows the gestalt principle of proximity, grouping the KPI summary at the top and arranging the charts in a logical flow from contract-level analysis to payment, demographics, tenure, and service type. The consistent teal and coral colour scheme applies the principle of similarity, making it easy to distinguish between retained and churned customers across all visuals. The Contract Type chart acts as a global filter, allowing the audience to click on any contract type and see all other charts update dynamically. This interactive capability transforms the dashboard from a static report into an exploratory tool that the audience can use to test hypotheses and drill into specific segments.

Further Exploration

Beyond the current dashboard, there are several additional questions that could be explored with this dataset. For instance, a heatmap or matrix visual could be used to examine churn rates across combinations of contract type and payment method, revealing interaction effects between these two variables. A treemap could visualise churn distribution across service add-ons like online security, tech support, and device protection, helping identify which bundled services are most effective at retaining customers. A scatter plot of monthly charges versus total charges, coloured by churn status, could reveal whether high-paying short-tenure customers

represent a distinct at-risk segment. These additional analyses would deepen the insights available to the retention team and could be incorporated into a multi-page Power BI report.

Conclusion

The Telecom Customer Churn Analysis dashboard tells a clear and actionable story. The typical churned customer is on a month-to-month contract, pays via electronic check, subscribes to fiber optic internet, and tends to leave within the first few months of their tenure. Gender does not play a meaningful role in predicting churn. The dashboard combines five distinct chart types - clustered bar, stacked bar, area chart, donut chart, and bar chart - each selected to match the nature of the underlying data. The KPI-first layout, consistent colour scheme, interactive filtering, and logical grouping of visuals all contribute to a dashboard that is visually appealing, easy to navigate, and directly addresses the research question. These insights can inform practical retention strategies such as incentivising longer contracts, addressing fiber optic service quality, promoting automatic payment methods, and investing in early-tenure customer engagement programs.

References

References

Power BI Tutorial. (2026, February 12). GeeksforGeeks. Retrieved March 13, 2026, from

https://www.geeksforgeeks.org/power-bi/power-bi-tutorial/

Telco Customer Churn. (n.d.). Kaggle. Retrieved March 13, 2026, from

https://www.kaggle.com/datasets/blastchar/telco-customer-churn